



Hybrid Techniques for the Fault Diagnosis of Mechanical Industrial Equipment

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Research Professor

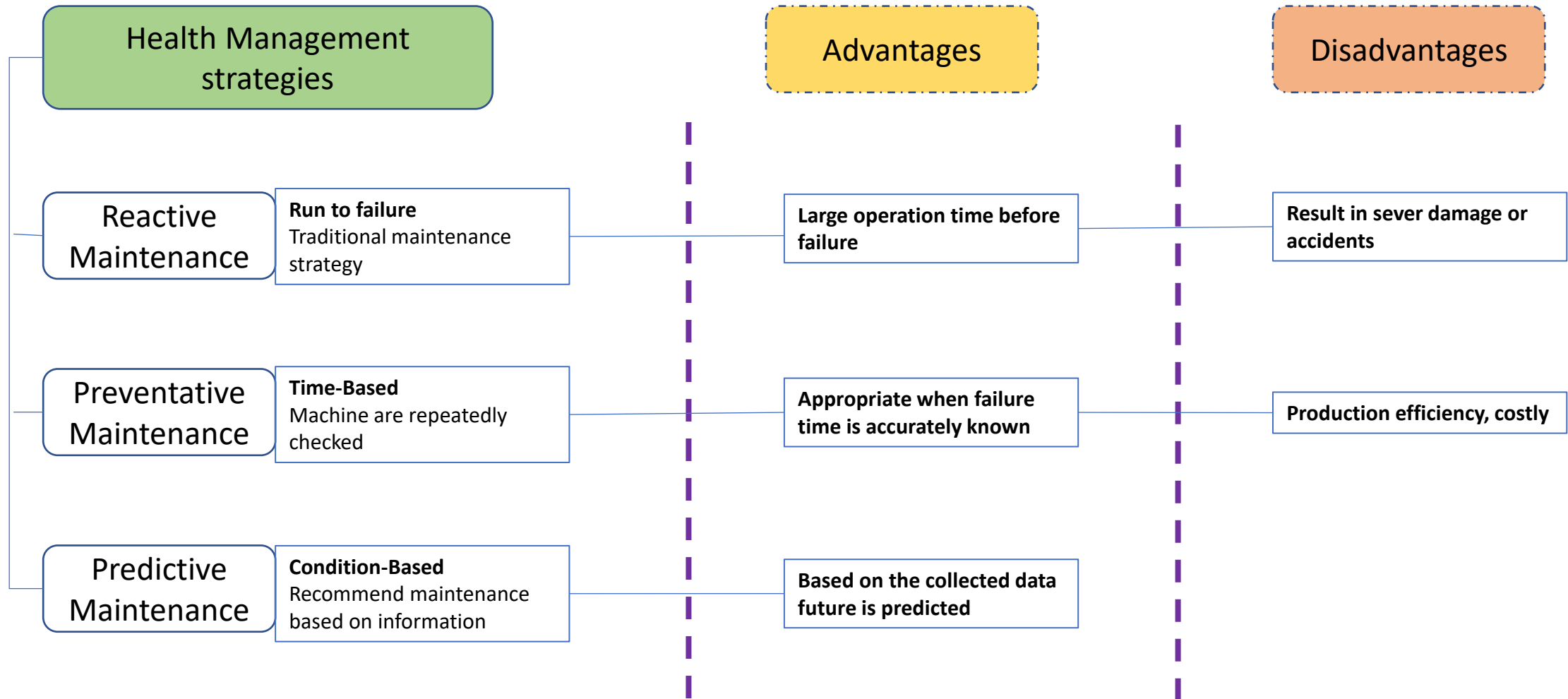
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Dept. Computer Engineering

2025-05-16



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How to Diagnose the Faults?



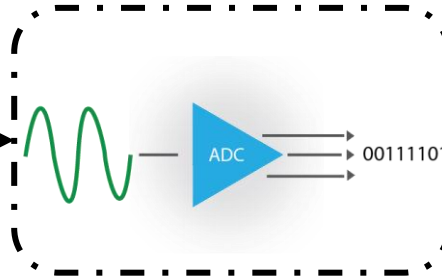
Reference: Lei, Yaguo. *Intelligent fault diagnosis and remaining useful life prediction of rotating machinery*. Butterworth-Heinemann, 2016.

Data Acquisition

Piezo-electric
accelerometer
(Adhesive)



Charge Amplifier



Analog to digital
converter

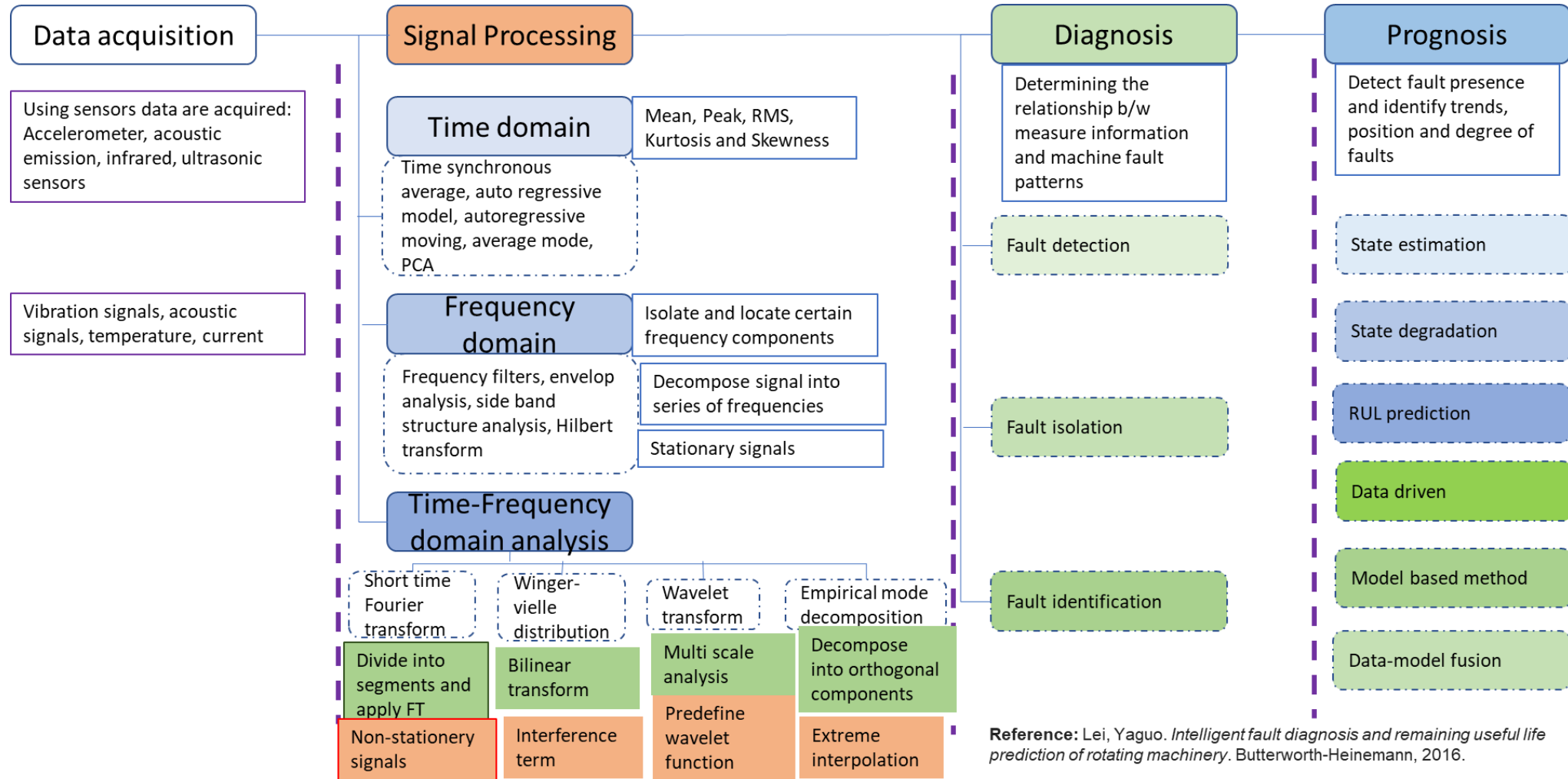


Single conditioning unit

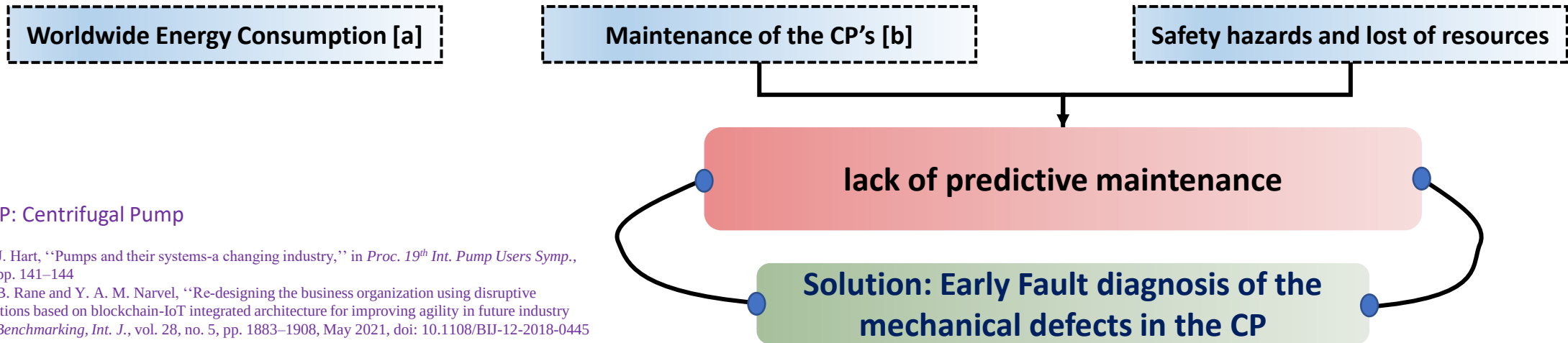
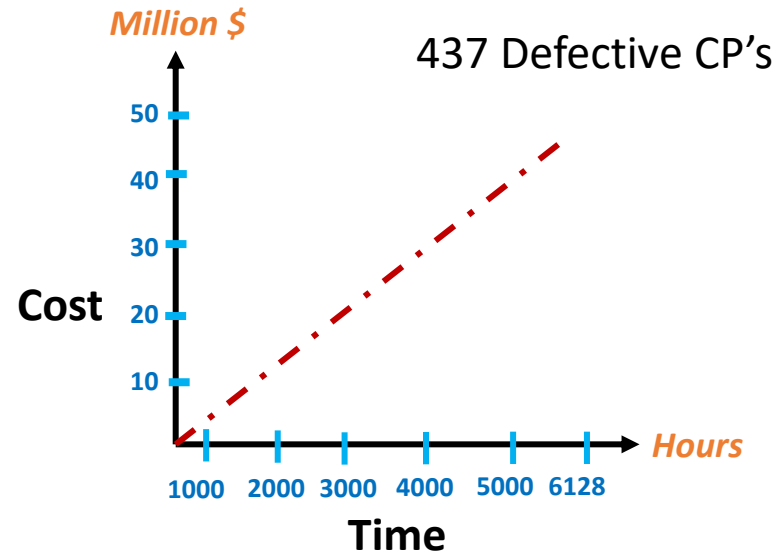
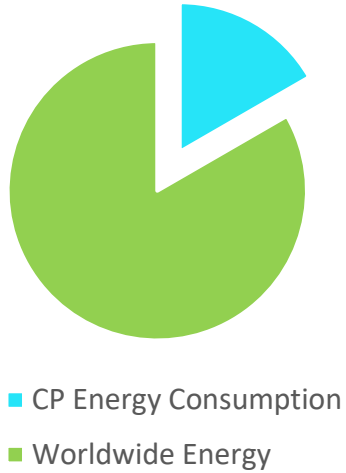


Storage

Condition Based Monitoring



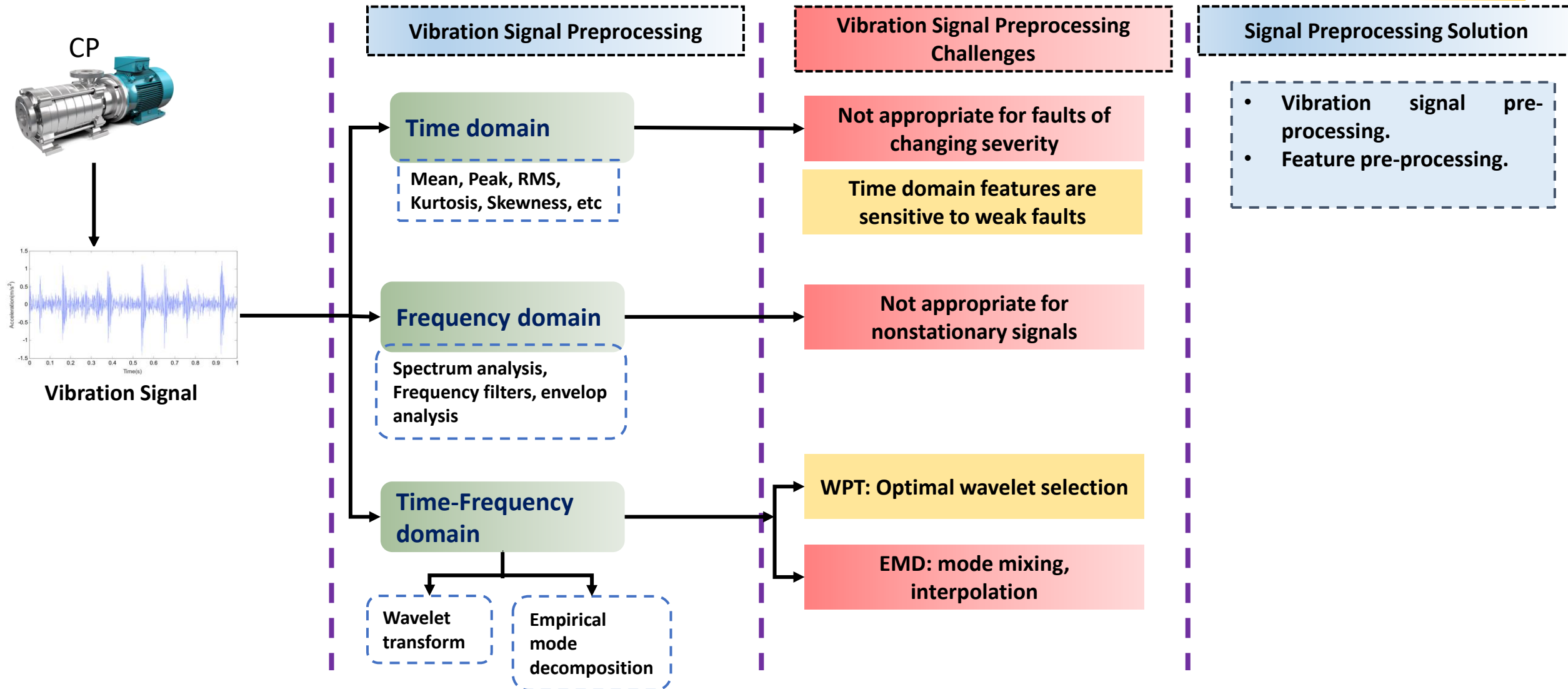
A Case study of Centrifugal Pump Fault Diagnosis



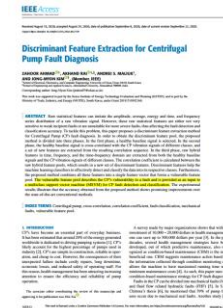
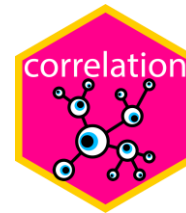
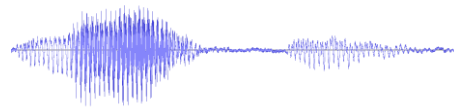
CP: Centrifugal Pump

- [a] R. J. Hart, "Pumps and their systems-a changing industry," in *Proc. 19th Int. Pump Users Symp.*, 2002, pp. 141–144
- [b] S. B. Rane and Y. A. M. Narvel, "Re-designing the business organization using disruptive innovations based on blockchain-IoT integrated architecture for improving agility in future industry 4.0," *Benchmarking, Int. J.*, vol. 28, no. 5, pp. 1883–1908, May 2021, doi: 10.1108/BIJ-12-2018-0445

Related Work and Challenges



IEEE Access



Discriminant Feature Extraction for Centrifugal Pump Fault Diagnosis

Impact Factor: 3.36

Motivation

CH-3

Discriminant Feature Extraction for Centrifugal Pump Fault Diagnosis

- 1 Research Problem, Objectives, and Proposal.
- 2 Proposed Method.
- 3 Results and Comparisons.

Research Problem, Objectives, and Proposal

Research Problem

- Traditional Features extracted from the CP vibration signal can affect the fault classification accuracy.
 - Not sensitive to incipient faults.
 - Not suitable for severe faults.

Objective

- Developing new set of discriminant features for CP fault diagnosis.

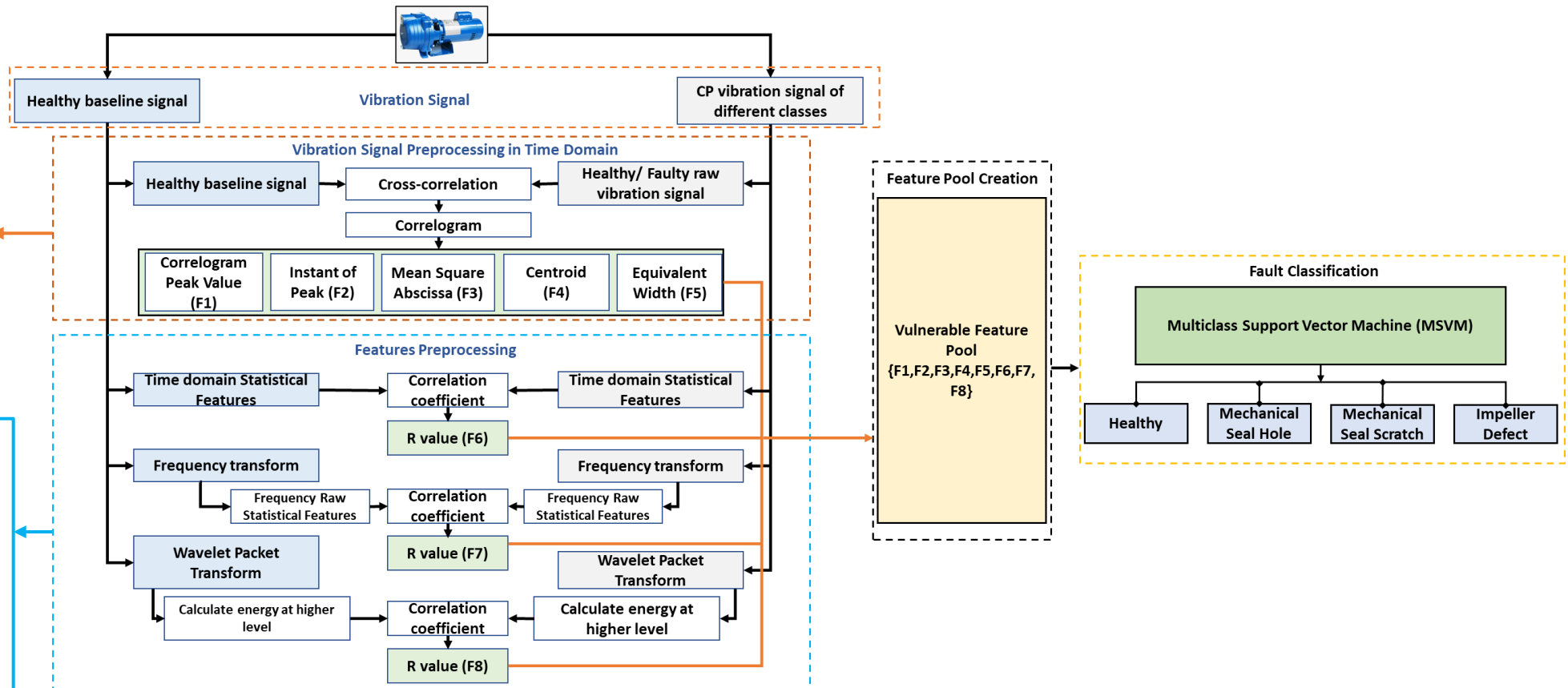
Proposal

- New set of features extraction from the correlation sequence obtained from the CP vibration signal preprocessing in the time domain.
- New set of features extraction from the preprocessing of raw statistical features based on correlation.

Proposed Method

The healthy baseline signal is cross-correlated with the CP vibration signal and a new set of discriminant features from the correlogram sequences

Raw statistical features are extracted from health baseline and CP vibration signals in time, frequency, and the time-frequency domain. This work then calculates the correlation coefficient between the two hybrid feature pools, which results in a new set of discriminant features for CP fault diagnosis.



<Figure: Proposed Method>

Proposed Method

Example

- Cross Correlation

$$R_{xy}[m] = \sum_{n=-\infty}^{\infty} x[n] y^*[n-m] \quad \cdots \text{Eq-1}$$

$x(t)$ and $y(t)$ are two signals as given in Eq-1.

$$x(t) = [-3, 2, -1, 1] \quad y(t) = [-1, 0, -3, 2]$$

			-3	2	-1	1			
-1	0	-3	2						
			-6						

$$R_{xy} = -6$$

			-3	2	-1	1			
	-1	0	-3	2					
			9	4					

$$R_{xy} = 13$$

			-3	2	-1	1			
		-1	0	-3	2				
			0	-6	-2				

$$R_{xy} = -8$$

			-3	2	-1	1			
			-1	0	-3	2			
			3	0	3	2			

$$R_{xy} = 8$$

			-3	2	-1	1			
				-1	0	-3	2		
				-2	0	-3			

$$R_{xy} = -5$$

			-3	2	-1	1			
					-1	0	-3	2	
					1	0			

$$R_{xy} = 1$$

			-3	2	-1	1			
						-1	0	-3	2
						-1			

$$R_{xy} = -1$$

Zeroth sample



$$R_{xy} = \{-6, 13, -8, 8, -5, 1, -1\}$$



Samples in the two signals are identical

Proposed Method: Healthy baseline Signal Selection

Pump Peak Efficiency Point (PEP): the point at which the pump operates at the highest efficiency.

$$\eta = \frac{Q\rho gH}{P} \quad \text{--- Eq. (a)}$$

Flow rate (Q):

The volume of fluid each second that is passing through a cross-sectional slice of the pump inlet.

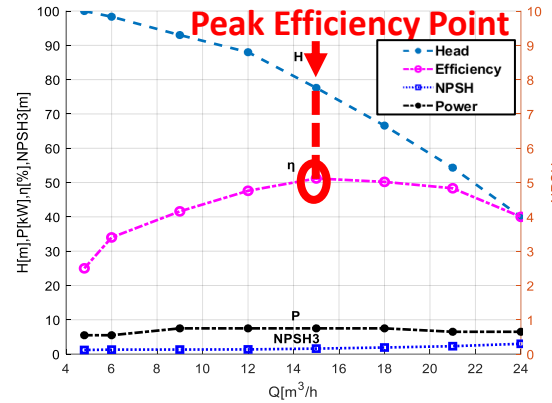
Head (H): the head is used to measure the kinetic energy which a pump creates.

NPSH:

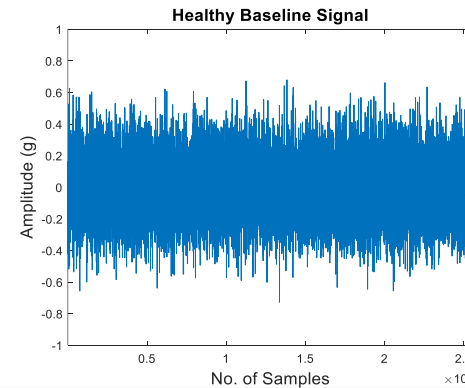
Net positive suction head

Power (P):

Pump coupling power consumption



<Figure: PMT-4008 Pump Characteristic Curve>



<Figure: Baseline Signal>



Collect signals at PEP for the 20s

Criteria

The signal mean should be near the vibration signal of normal condition and furthest from the defective operating condition.

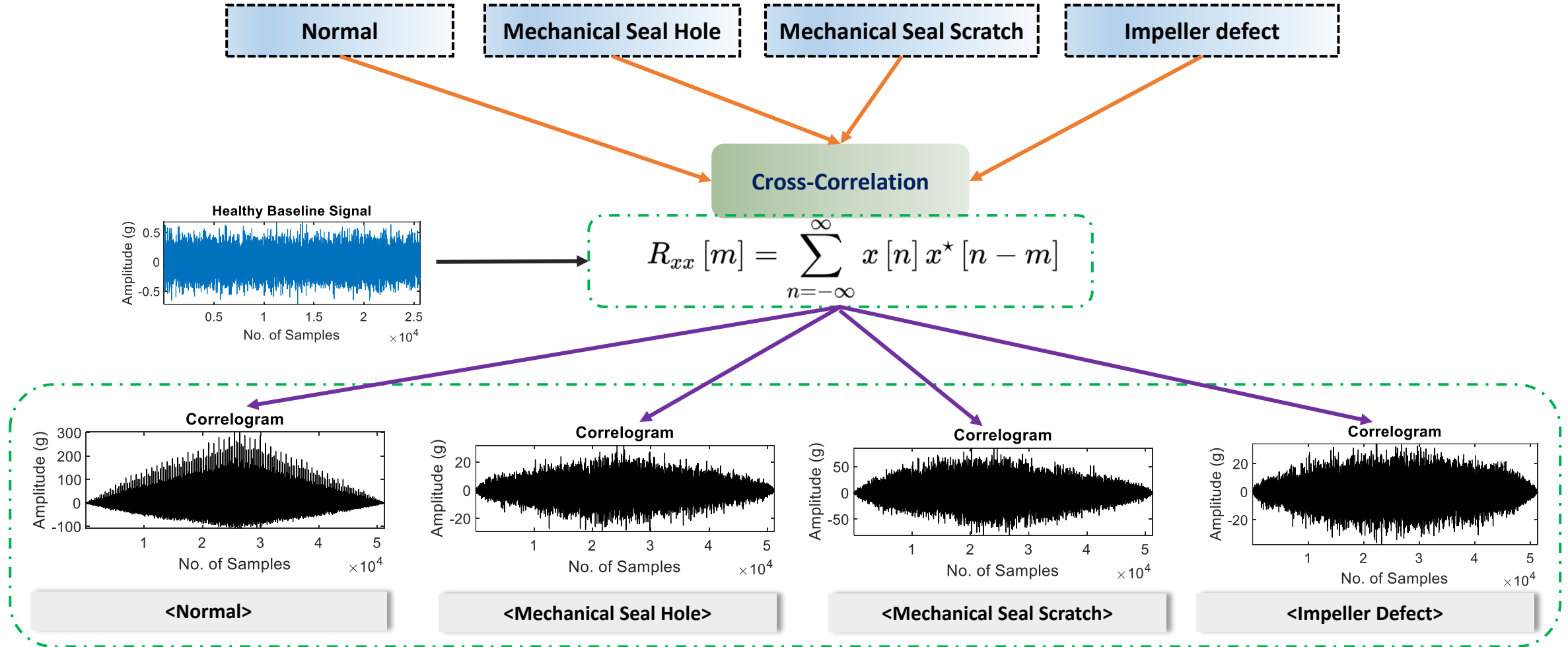
Yes

Discard

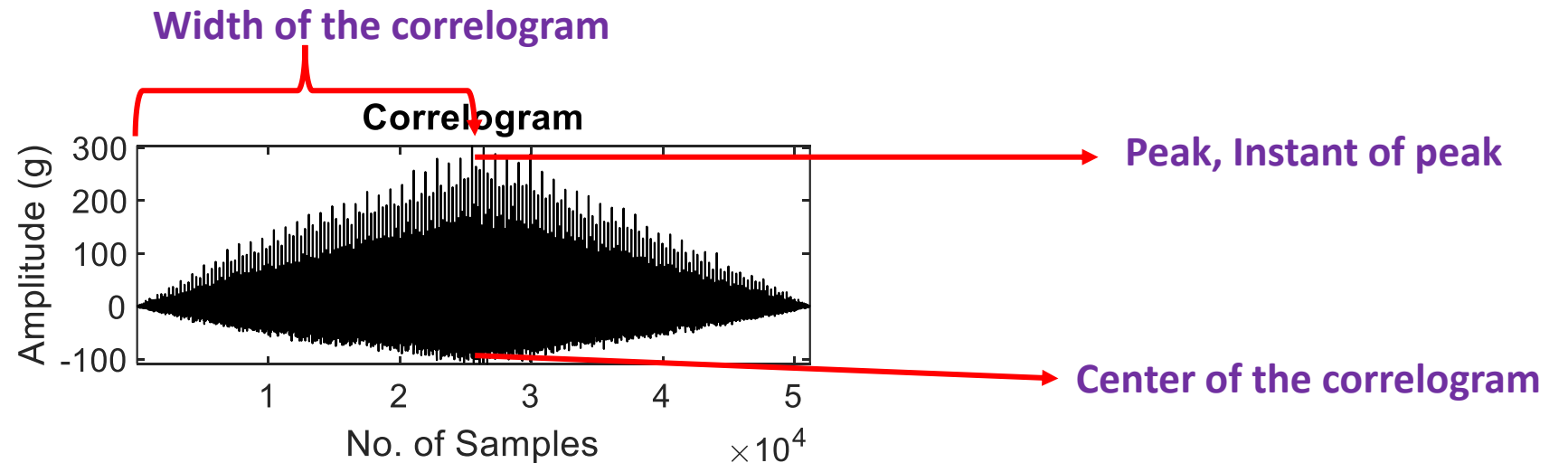
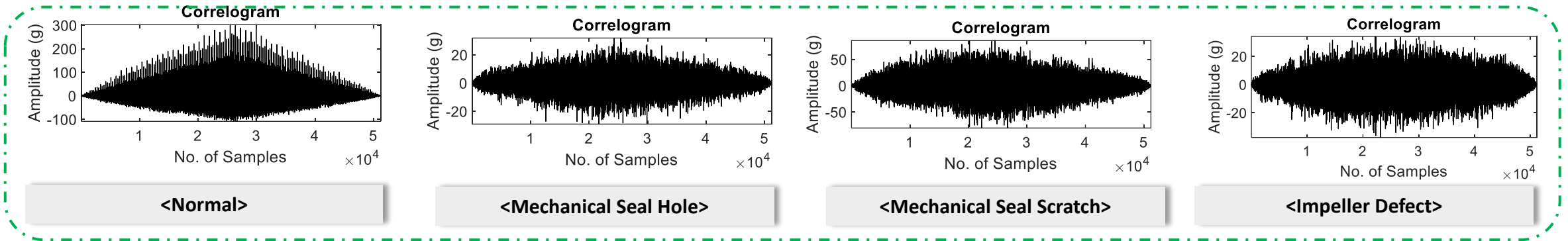
Select

Healthy baseline signal

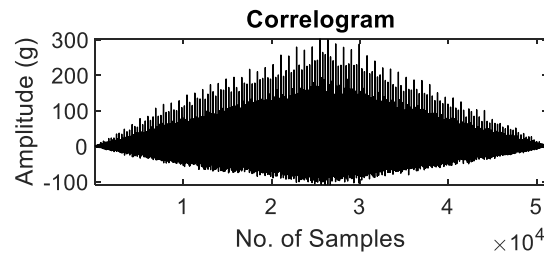
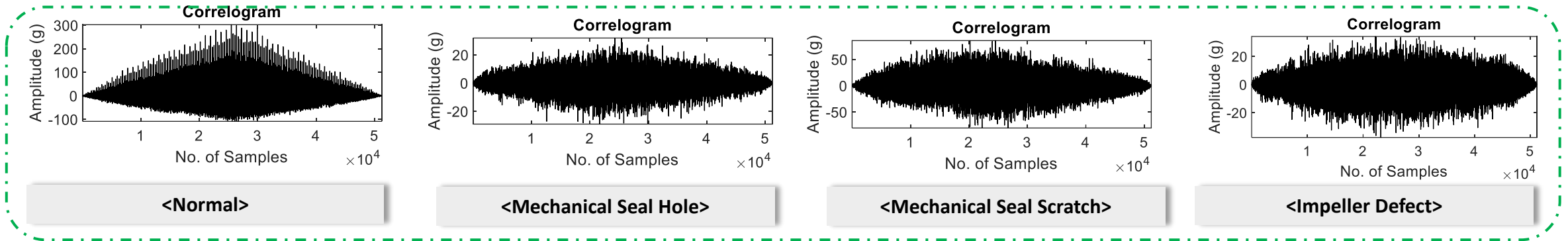
Proposed Method: Time Domain Signal Preprocessing



Proposed Method: Time Domain Signal Preprocessing



Proposed Method: Time Domain Signal Preprocessing



R_m = Correlogram sequence
 m = the time shift

Correlogram Centroid

$$= \frac{\sum_{m=-M}^M mR(m)}{\sum_{m=-M}^M R(m)}$$

Equivalent Width

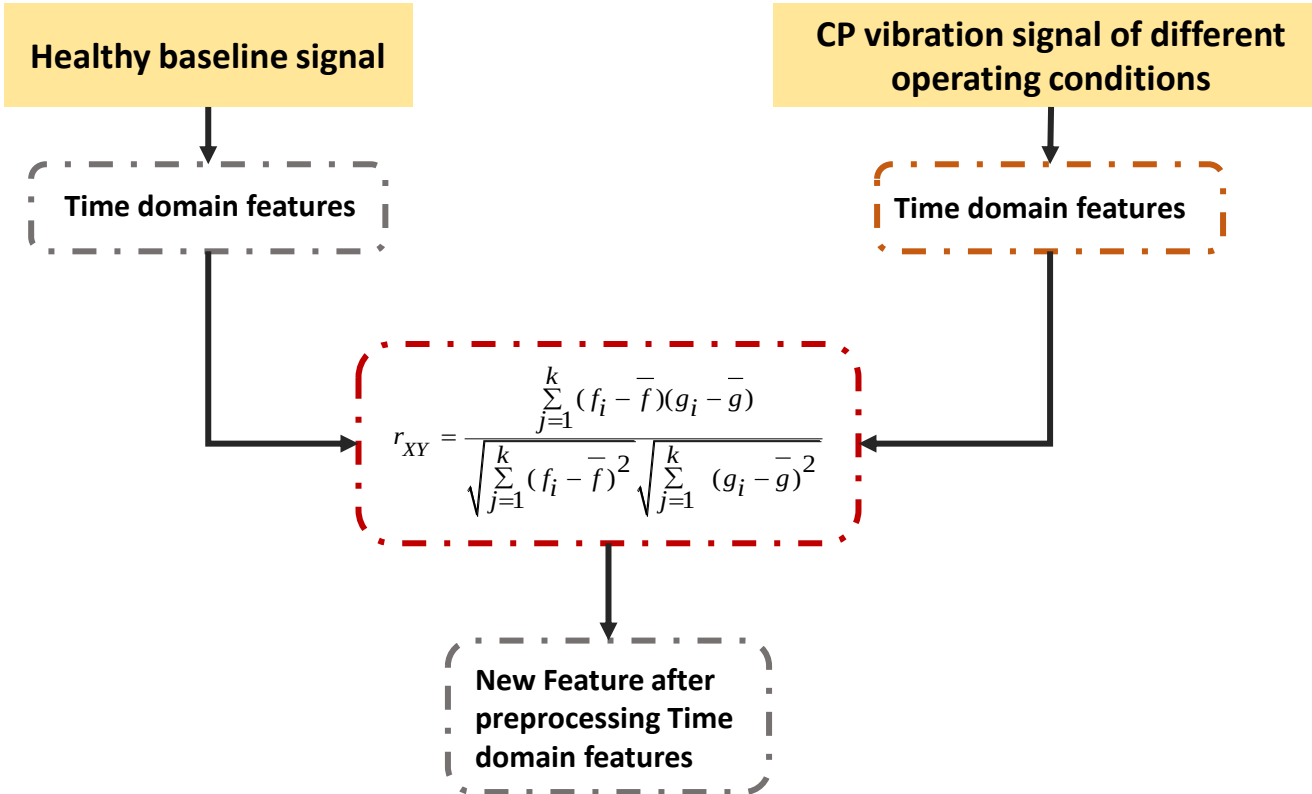
$$= \frac{\sum_{m=-M}^M mR(m)}{\text{Peak value of } R(m)}$$

Mean square Abscissa

$$= \frac{\sum_{m=-M}^M m^2 R(m)}{\sum_{m=-M}^M R(m)}$$

Proposed Method: Feature Preprocessing

The random fluctuations caused by multiple severity faults in the CP cannot be addressed by analysing the signal only in the time or frequency or time-frequency domain.



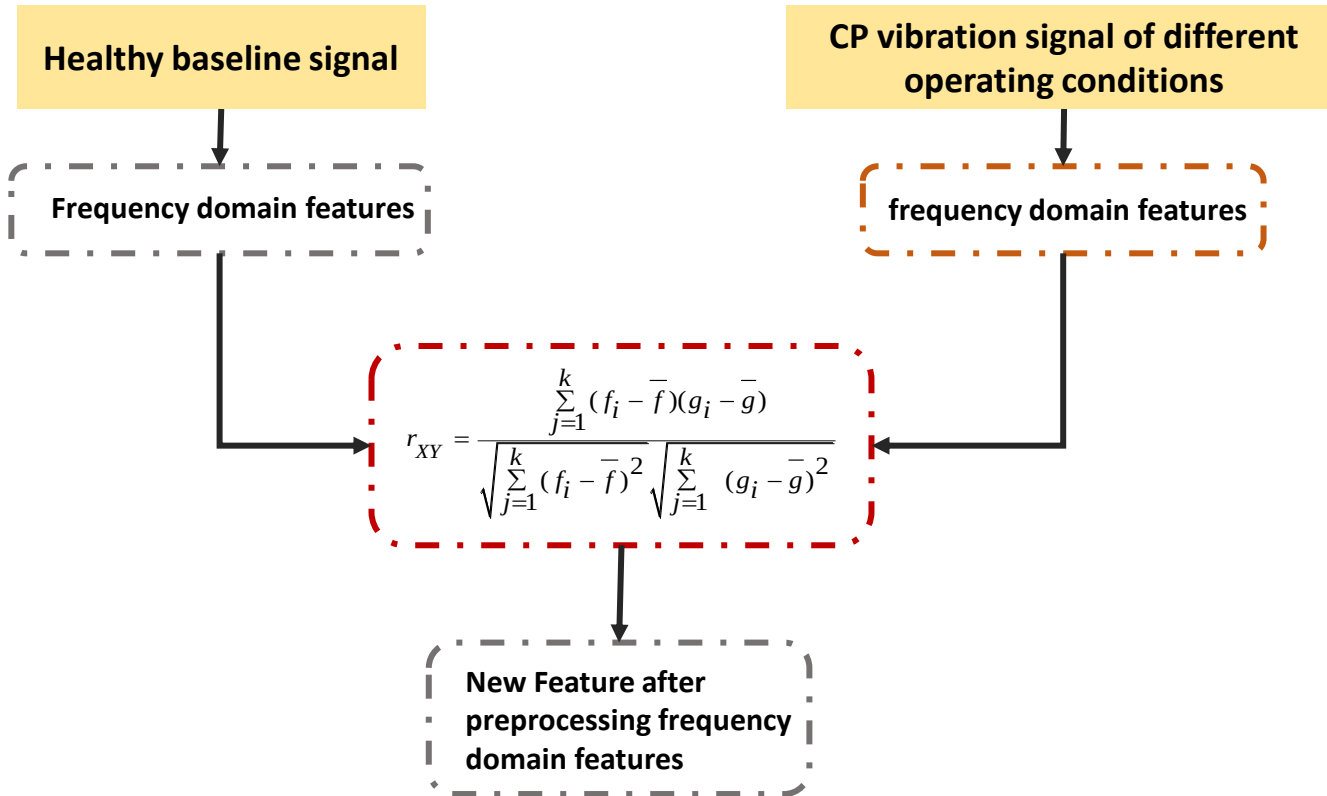
f, g = are the two feature vectors

Extracted Features: peak, mean, variance, standard deviation, kurtosis, impulse factor, shape factor, clearance factor, crest factor, root mean square, root factor, and skewness.

Feature	Equation	Feature	Equation	Feature	Equation
Mean	$X_m = \frac{\sum_{n=1}^N x(n)}{N}$	Root amplitude	$X_{root} = \left[\frac{\sum_{n=1}^N \sqrt{ x(n) }}{N} \right]^2$	Peak	$X_{peak} = \max x(n) $
Standard deviation	$X_{sd} = \sqrt{\frac{\sum_{n=1}^N (x(n) - X_m)^2}{N-1}}$	Root mean square	$X_{rms} = \sqrt{\frac{\sum_{n=1}^N (x(n))^2}{N}}$	Skewness	$X_{sk} = \frac{\sum_{n=1}^N (x(n) - X_m)^3}{(N-1)X_{sd}^3}$
Kurtosis	$X_{kurtosis} = \frac{\sum_{n=1}^N (x(n) - X_m)^4}{(N-1)X_{sd}^4}$	Crest factor	$X_{crest} = \frac{X_{peak}}{X_{rms}}$	Clearance factor	$X_{clearance} = \frac{X_{peak}}{X_{root}}$
Shape factor	$X_{shape} = \frac{X_{rms}}{\frac{1}{N} \sum_{n=1}^N x(n) }$	Impulse factor	$X_{impulse} = \frac{X_{peak}}{\frac{1}{N} \sum_{n=1}^N x(n) }$		

$X(n)$ = time domain signal

Proposed Method: Feature Preprocessing



f, g = are the two feature vectors

Extracted Features: Mean frequency, root mean frequency, root mean square frequency, standard deviation, spectral kurtosis,

Feature	Equation	Feature	Equation	Feature	Equation
Mean Frequency	$X_{mf} = \frac{\sum_{k=1}^K s(k)}{K}$	Root variance frequency	$X_{rvf} = \sqrt{\frac{\sum_{k=1}^K (s(k) - X_{mf})^2}{K}}$	Root mean square frequency	$X_{frms} = \sqrt{\frac{\sum_{k=1}^K (s(k))^2}{K}}$
Standard deviation	$\sigma_f^2 = \frac{\sum_{k=1}^K (s(k) - X_{mf})^2}{K - 1}$	Spectral Kurtosis	$X_{fkurtosis} = \frac{\sum_{k=1}^K (s(k) - X_{mf})^4}{(K)\sigma_f^4}$		

$s(k)$ = Frequency spectrum

Proposed Method: Feature Preprocessing

Healthy baseline signal

Time-Frequency domain features

CP vibration signal of different operating conditions

Time-frequency domain features

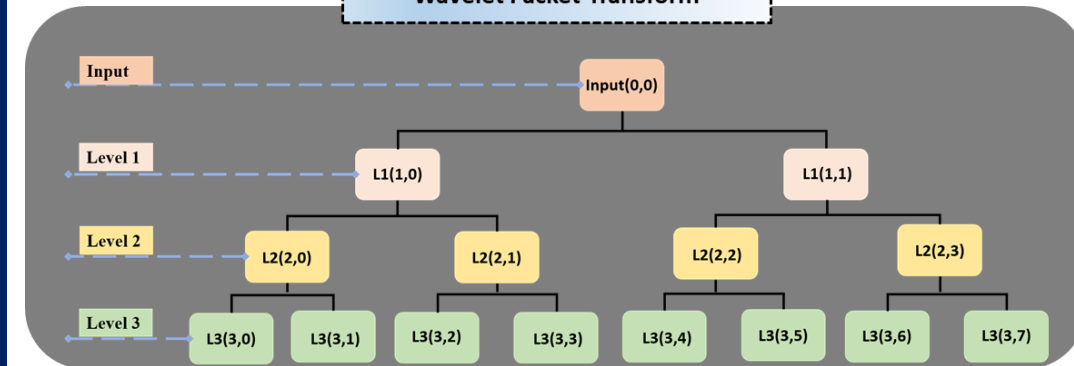
$$r_{XY} = \frac{\sum_{j=1}^k (f_j - \bar{f})(g_j - \bar{g})}{\sqrt{\sum_{j=1}^k (f_j - \bar{f})^2} \sqrt{\sum_{j=1}^k (g_j - \bar{g})^2}}$$

New Feature after preprocessing time-frequency domain features

f, g = are the two feature vectors

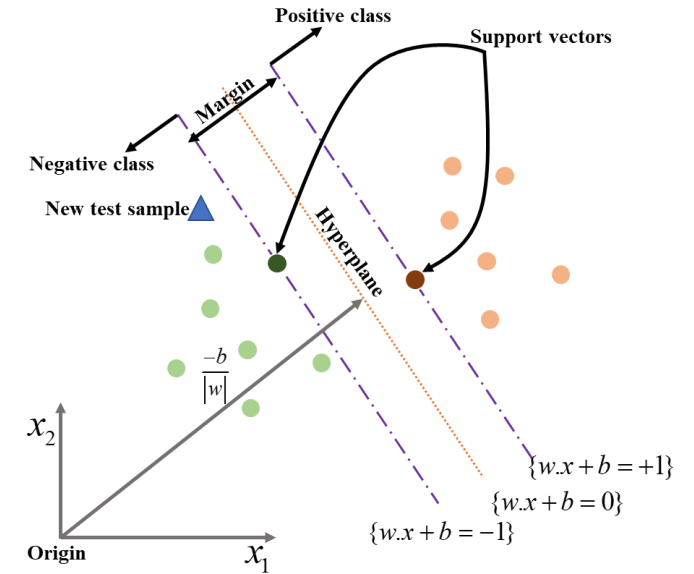
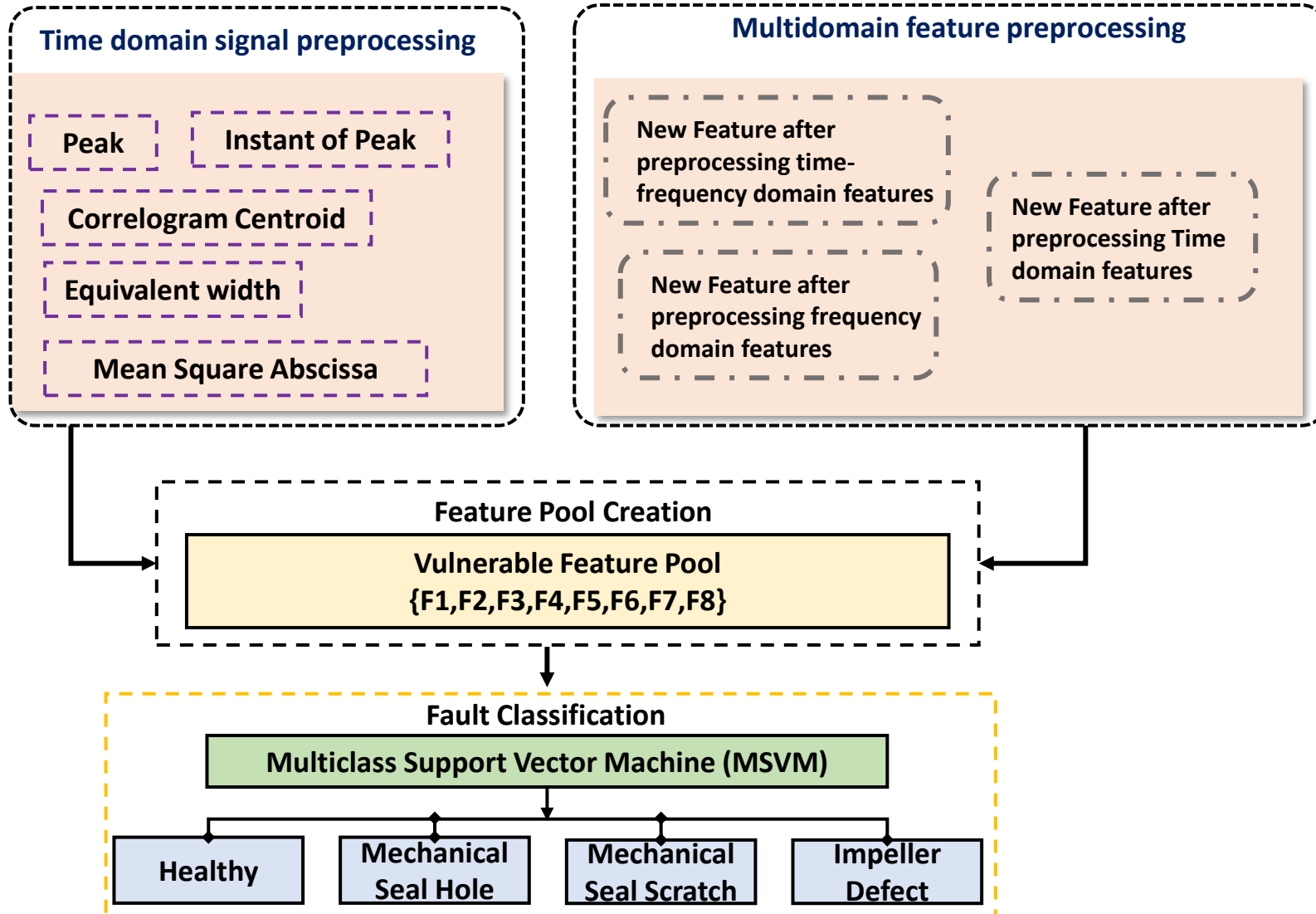
Wavelet: Daubechies 10
Decomposition Level: 3

Wavelet Packet Transform



From each frequency band energy is calculated as a feature.

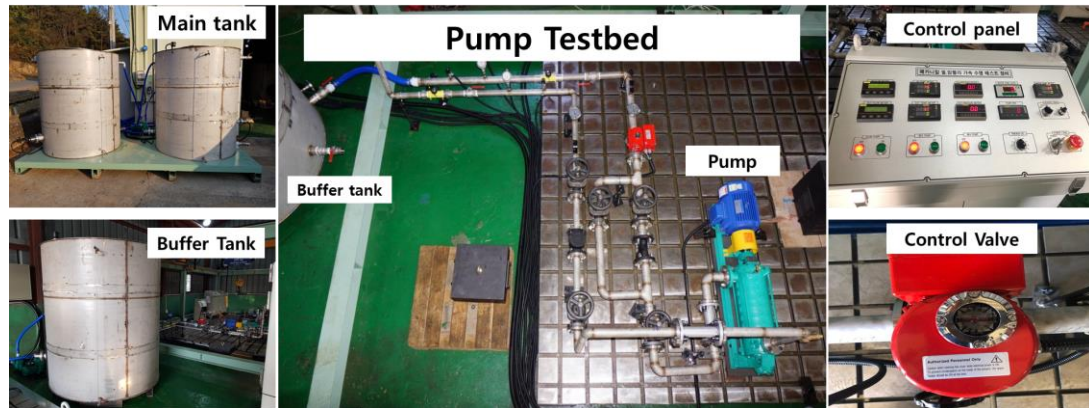
Proposed Method: Vulnerable Feature Pool



w = wights; x = variable; b = bias.

- One against all.
- Radial basis function kernel.
- Regularization $C = 1$.
- Gamma = 0.1 (using grid search method)

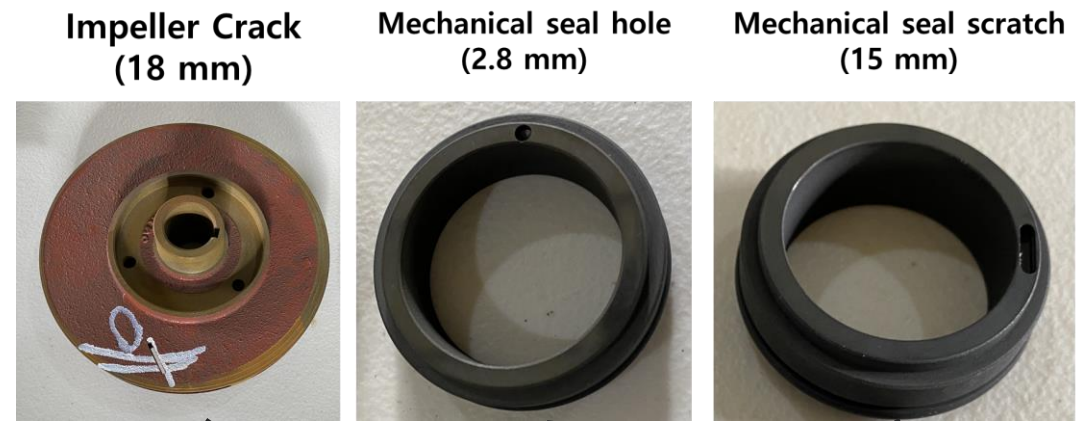
Dataset for Validation



<Figure: Experimental Setup>



<Figure: Accelerometer location>

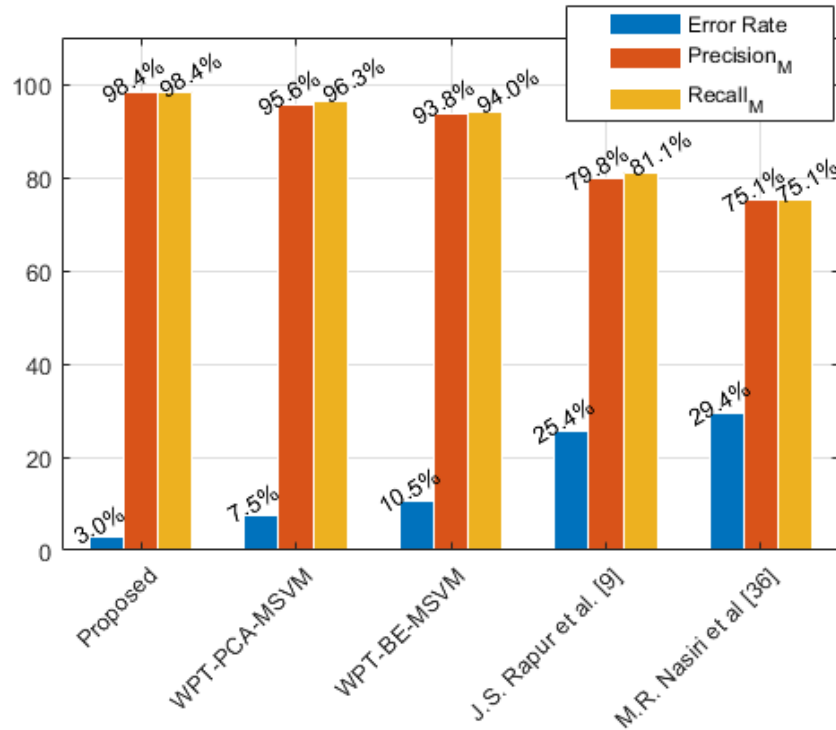


For each condition	Pressure
300 samples 300x4 =1200	{3} bar

<Figure: Dataset Description>

Results and Comparisons

The experiment has been performed 20 times. In each experimental trail, the training and testing sets were randomly sampled at a ratio of 8:2.



Methods	TPR in Average (%)				ACA (%)
	Normal Condition	Mechanical seal defect (hole)	Mechanical seal defect (scratch)	Impeller fault	
Proposed	100	96.2	98.9	98.4	98.4
WPT-PCA-MSVM	100	97.72	98.8	88.6	96.3
WPT-BE-MSVM	100	91.2	90.2	94.5	94
Rapur et al. [9]	68.9	83.6	91.5	80.4	81.1
Nasiri et al [36]	82.1	68.3	70.5	79.7	75.22

Average classification accuracy = 98.4%

$$T - PR = \frac{1}{k} \sum_{j=1}^k \left(\frac{(N_{TP}^{j,m})}{(N_{TP}^{j,m} + N_{FN}^{j,m})} \right) \times 100(\%)$$

$$E_r = \frac{1}{k} \sum_{j=1}^k \left(\frac{(N_{TP}^{j,m}) + (N_{FN}^{j,m})}{(N_{TP}^{j,m}) + (N_{FN}^{j,m}) + N_{TN}^{j,m} + N_{FP}^{j,m}} \right) \times 100(\%)$$

$$AAC = \frac{1}{k} \sum_{j=1}^k \left(\sum_{m=1}^L \frac{N_{TP}^{j,m}}{N_{samples}} \right) \times 100(\%)$$

True positive False negatives True Negatives False positive

[*] J. S. Rapur and R. Tiwari, "Experimental time-domain vibration-based fault diagnosis of centrifugal pumps using support vector machine," ASCEASME J Risk Uncertainty Eng. Syst. Part B, Mech. Eng., vol. 3, no. 4, pp. 1–7, Dec. 2017, doi: 10.1115/1.4035440.

[**] M. R. Nasiri, M. J. Mahjoob, and H. Vahid-Alizadeh, "Vibration signature analysis for detecting cavitation in centrifugal pumps using neural networks," in Proc. IEEE Int. Conf. Mechatronics, Apr. 2011, pp. 632–635, doi: 10.1109/ICMECH.2011.5971192.

Results and Comparisons

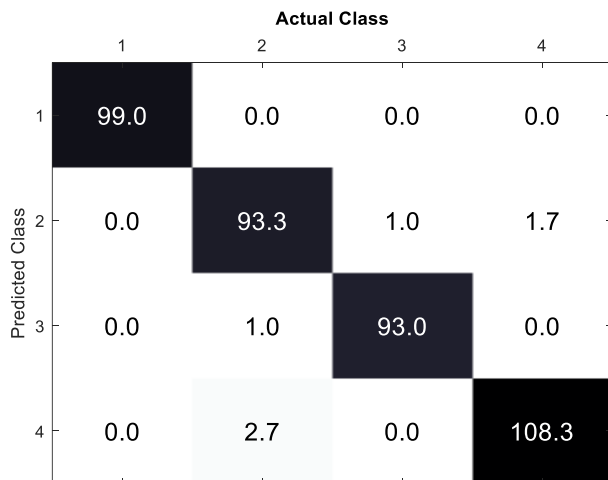
Proposed: good fault detection and classification accuracy with **few misclassifications** for mechanical seal hole defects.

WPT-PCA-MSVM: good fault detection and classification accuracy with **some misclassifications** for Impeller defects.

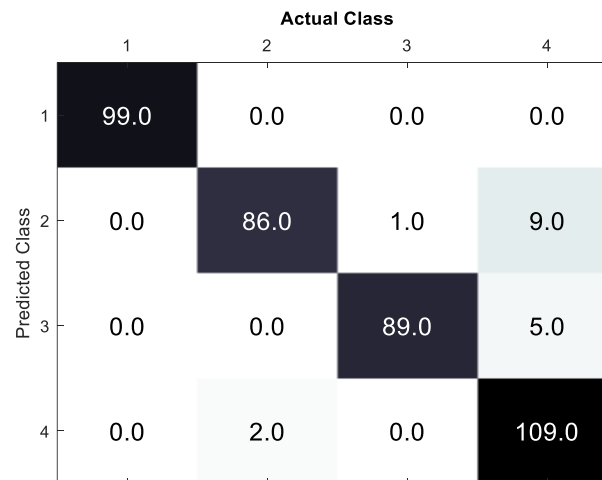
WPT-BE-MSVM: good fault detection and classification accuracy with **some misclassifications** for incipient faults such as mechanical seal holes and scratches.

Labels

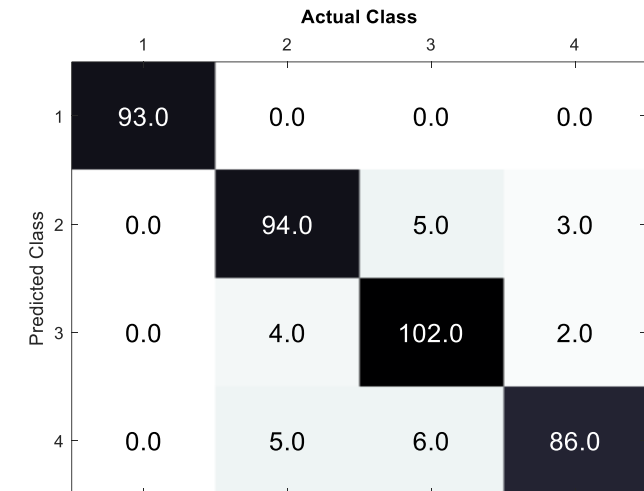
1: Healthy 2: Mechanical seal hole 3: Mechanical seal scratch 4: Impeller defect



<Figure: Proposed Method>



<Figure: WPT-PCA-MSVM>



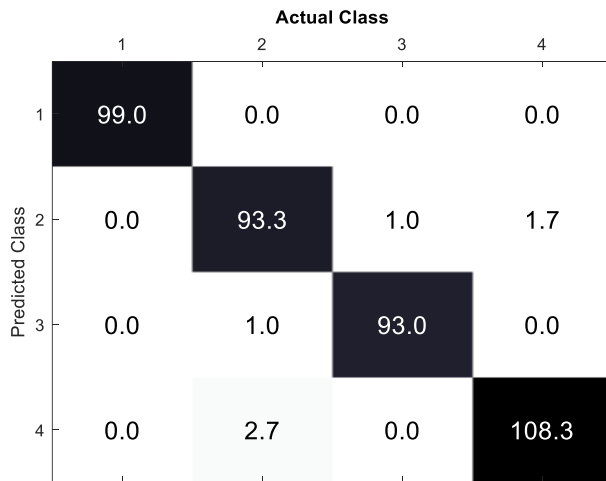
<Figure: WPT-BE-MSVM>

Results and Comparisons

Proposed: good fault detection and classification accuracy with **few misclassifications for mechanical seal hole defects.**
J. s. Rapur et al. [*]: bad fault detection and classification accuracy.
M. R. Nasiri et al. []:** bad fault detection and classification accuracy.

Labels

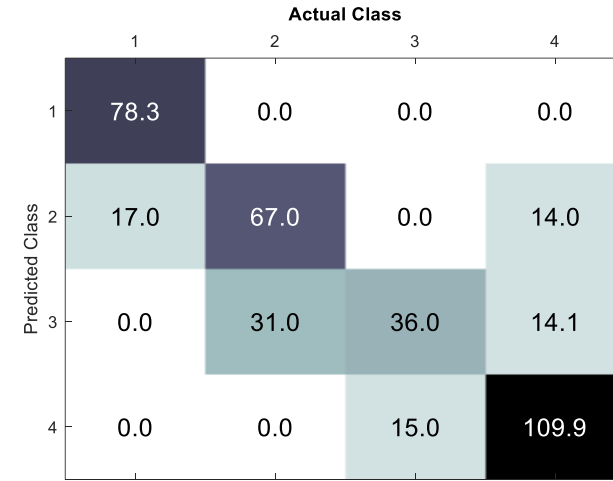
1: Healthy 2: Mechanical seal hole 3: Mechanical seal scratch 4: Impeller defect



<Figure: Proposed Method>



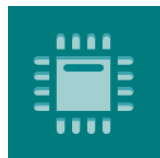
<Figure: J. s. Rapur et al. [*]>



<Figure: M. R. Nasiri et al. [**] >

[*] J. S. Rapur and R. Tiwari, "Experimental time-domain vibration-based fault diagnosis of centrifugal pumps using support vector machine," ASCEASME J Risk Uncertainty Eng. Syst. Part B, Mech. Eng., vol. 3, no. 4, pp. 1–7, Dec. 2017, doi: 10.1115/1.4035440.

[**] M. R. Nasiri, M. J. Mahjoob, and H. Vahid-Alizadeh, "Vibration signature analysis for detecting cavitation in centrifugal pumps using neural networks," in Proc. IEEE Int. Conf. Mechatronics, Apr. 2011, pp. 632–635, doi: 10.1109/ICMECH.2011.5971192.



sensors



Multistage Centrifugal Pump Fault Diagnosis Using Informative Ratio Principal Component Analysis

Impact Factor: 3.57

Motivation

CH-4

Multistage Centrifugal Pump Fault Diagnosis Using Informative Ratio Principal Component Analysis

- 1 Research Problem, Objectives, and Proposal.
- 2 Proposed Method.
- 3 Results and Comparisons.

Research Problem, Objectives, and Proposal

Research Problem

- The interference background noise in the signal affects the features' sensitivity towards the faults.
- All the features extracted from the vibration signal are not appropriate for representing the centrifugal pump's health condition.

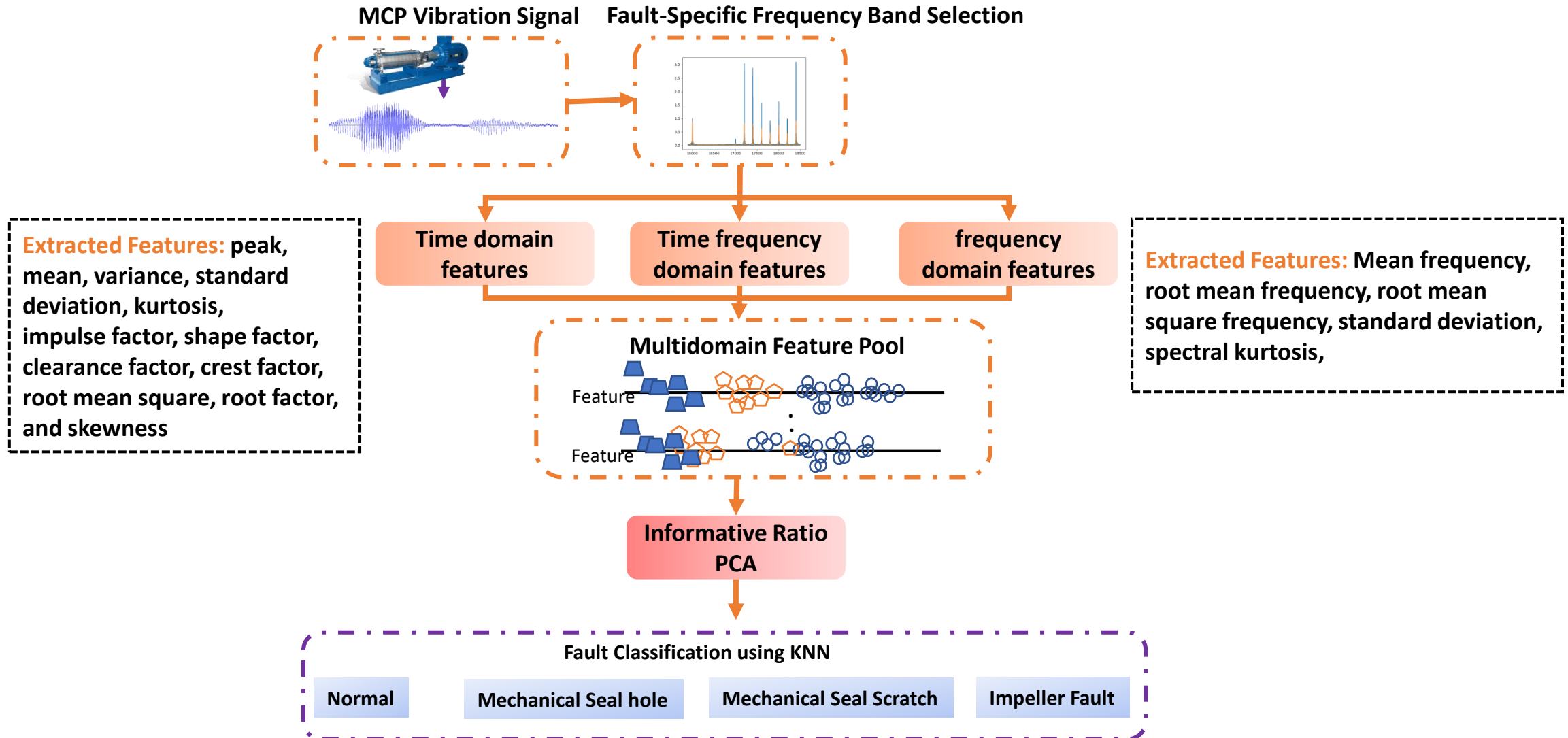
Objective

- A lightweight framework that will focus on the fault-specific frequencies and fault-sensitive features.

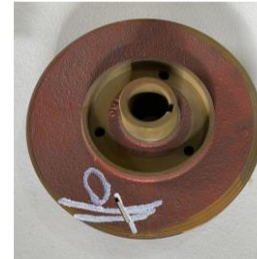
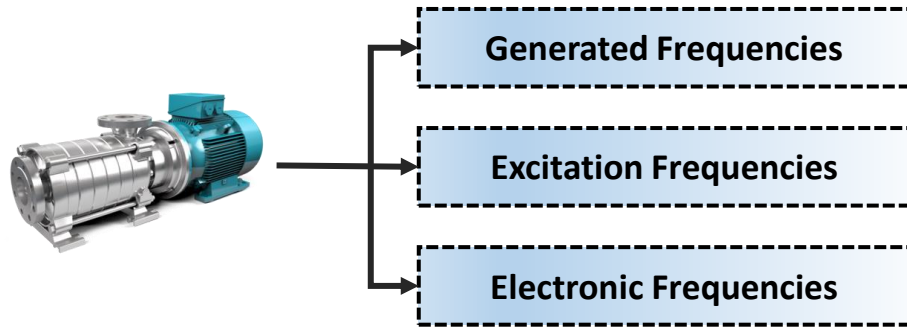
Proposal

- Selection of Fault specific frequency band.
- Novel Informative Ratio Principal Component Analysis for a new set of CP health-sensitive features selection.

Proposed Method



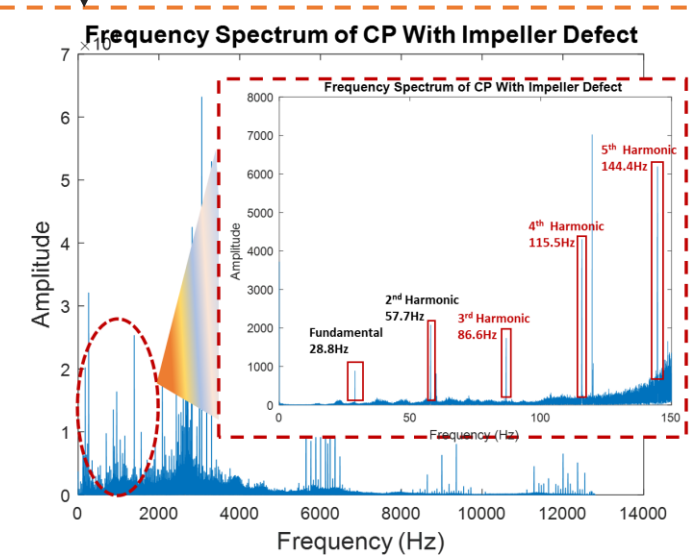
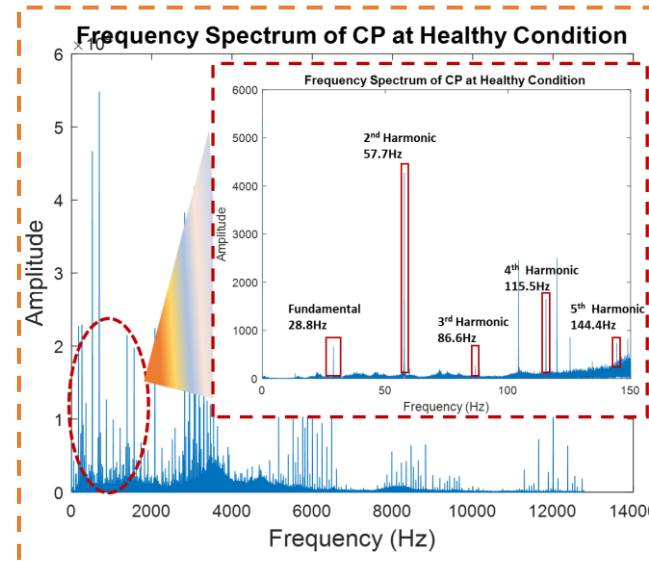
Proposed Method: Fault-Specific Frequency Band Selection



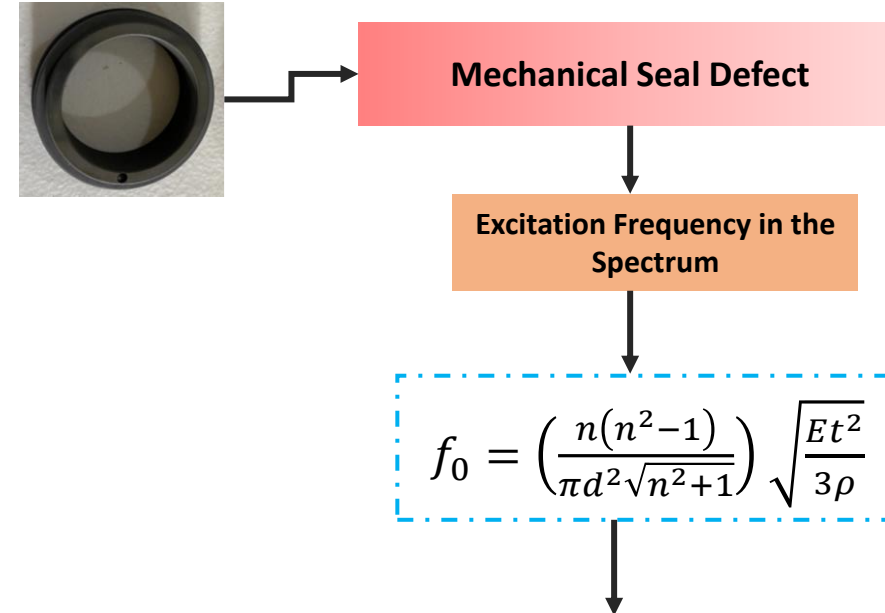
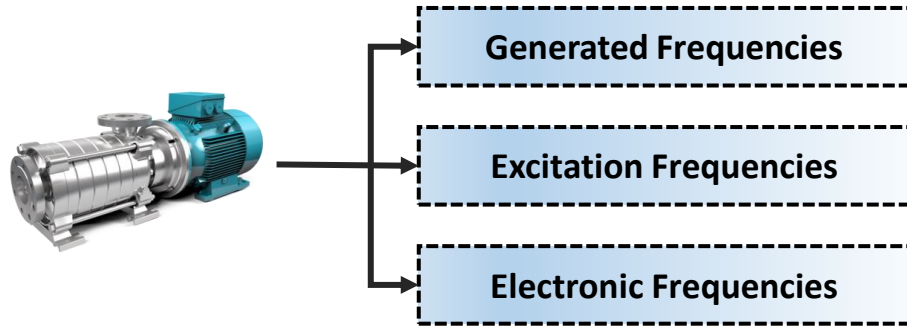
Impeller defect causes imbalance

Generated Frequencies in the Spectrum

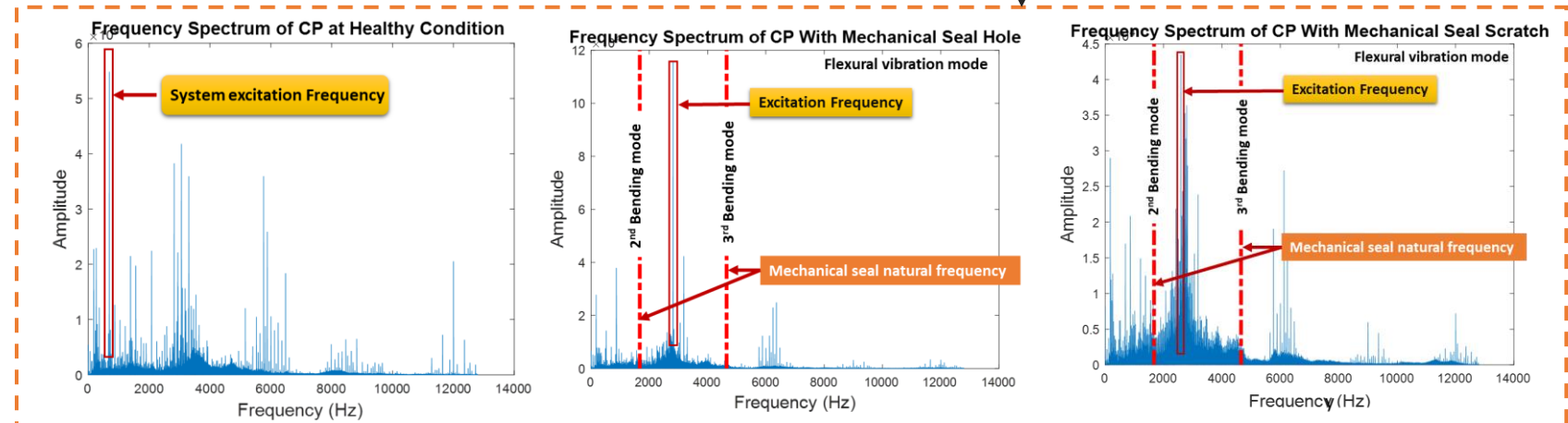
$$F_{i.defect} = z.N.$$



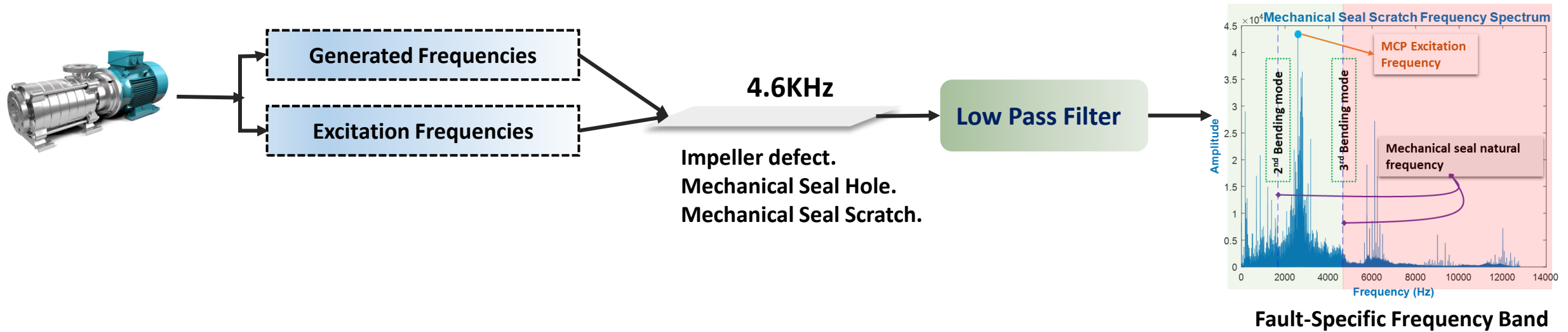
Proposed Method: Fault-Specific Frequency Band Selection



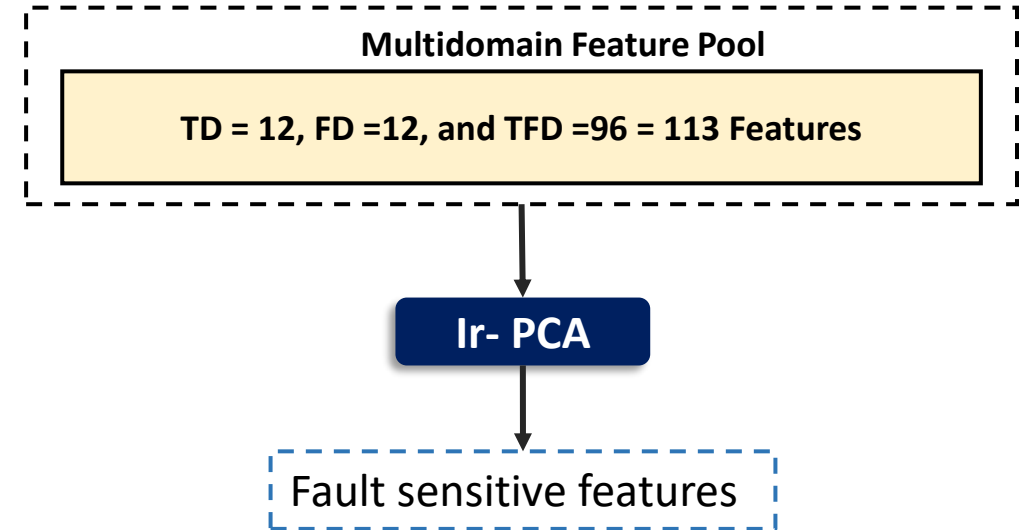
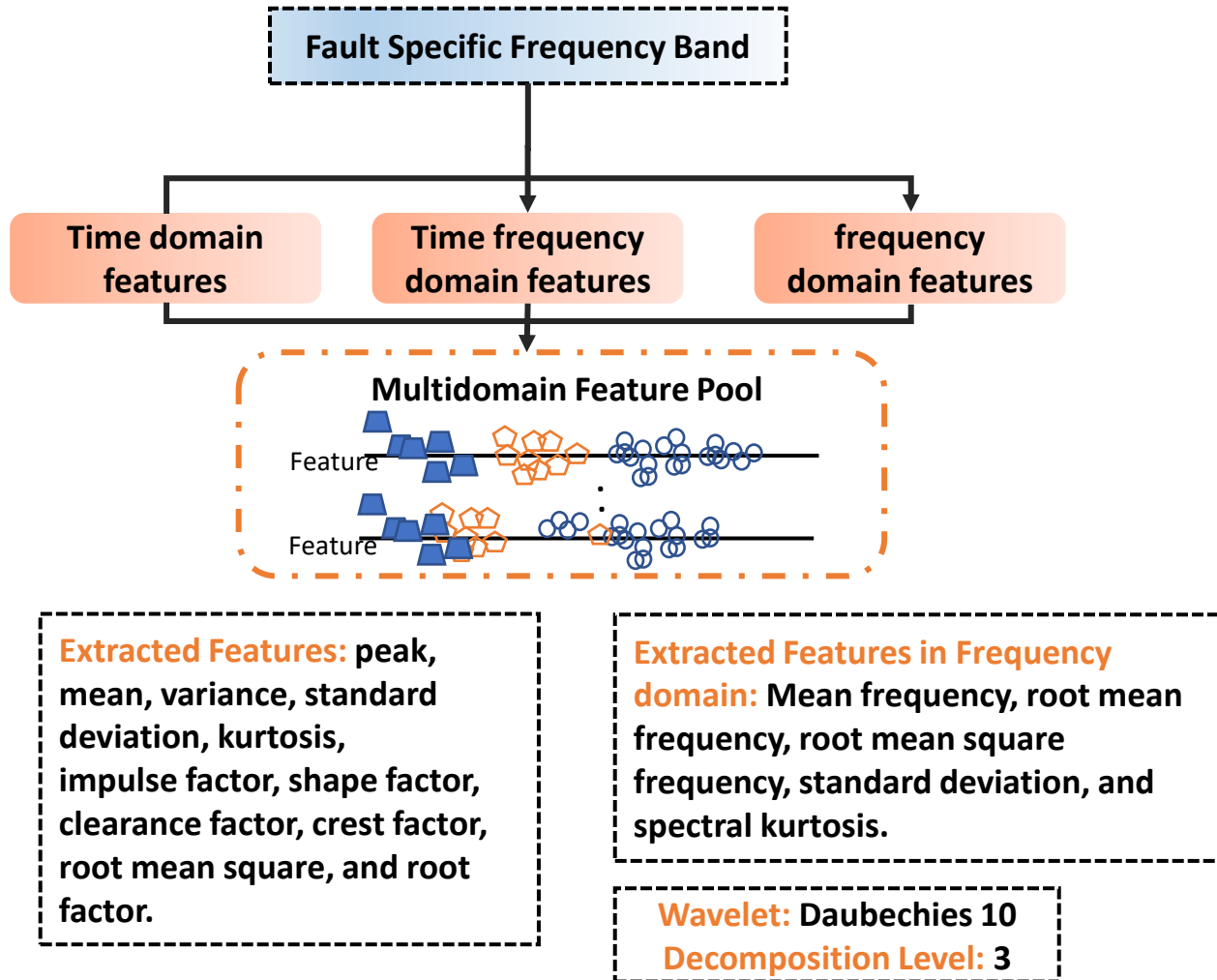
n = higher vibration modes = 1,2,3,4,5
 E : is modules of elasticity
 P : is the material density
 d = ring diameter
 t = ring cross-sectional thickness



Proposed Method: Fault-Specific Frequency Band Selection



Proposed Method: Feature Extraction

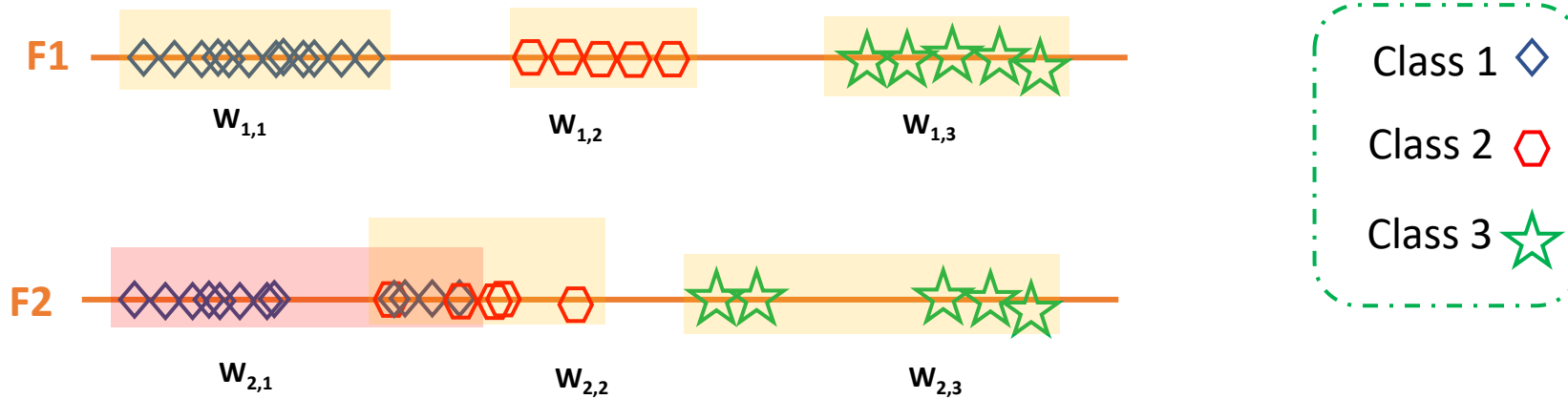


Proposed Method: Ir-PCA

Step-1: Within the class feature scatteredness.

$$I_i^S = \frac{1}{N} \sum_{n=1}^N I_{n,i}$$

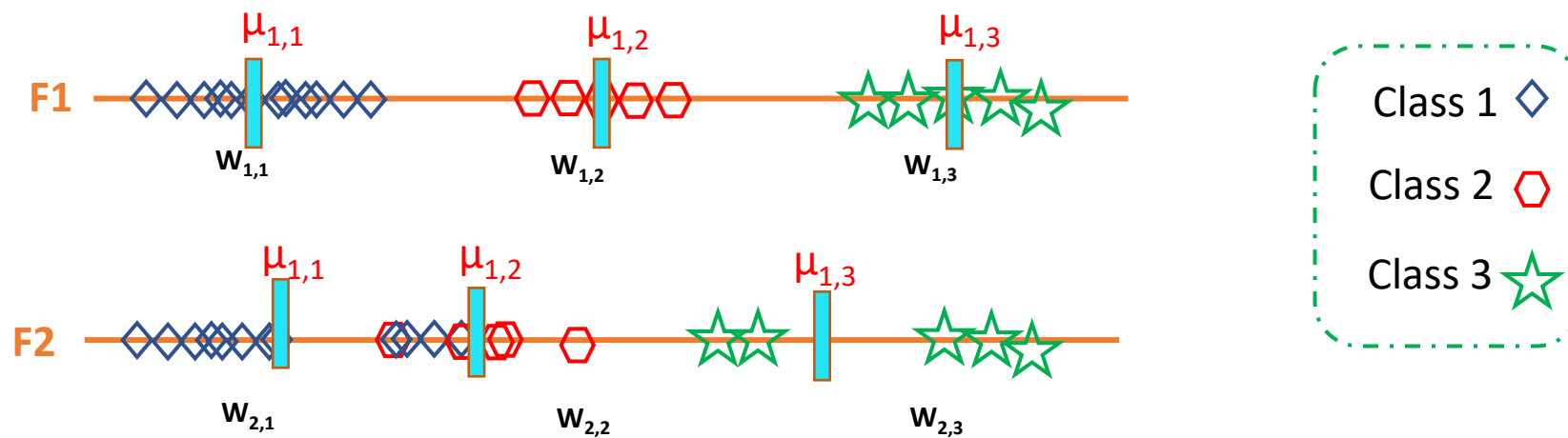
$$I_{n,i} = \frac{1}{S_n \times (S_n - 1)} \sum_{j,s=1}^{S_n} |x_{s,n,i} - x_{j,n,i}|, \text{ where } j, s = 1, 2, \dots, S_n, j \neq s.$$



Where N represents the classes, i is the number of features, x represents the feature, S is the sample number, and I_i^S is the inter class.

Proposed Method: Ir-PCA

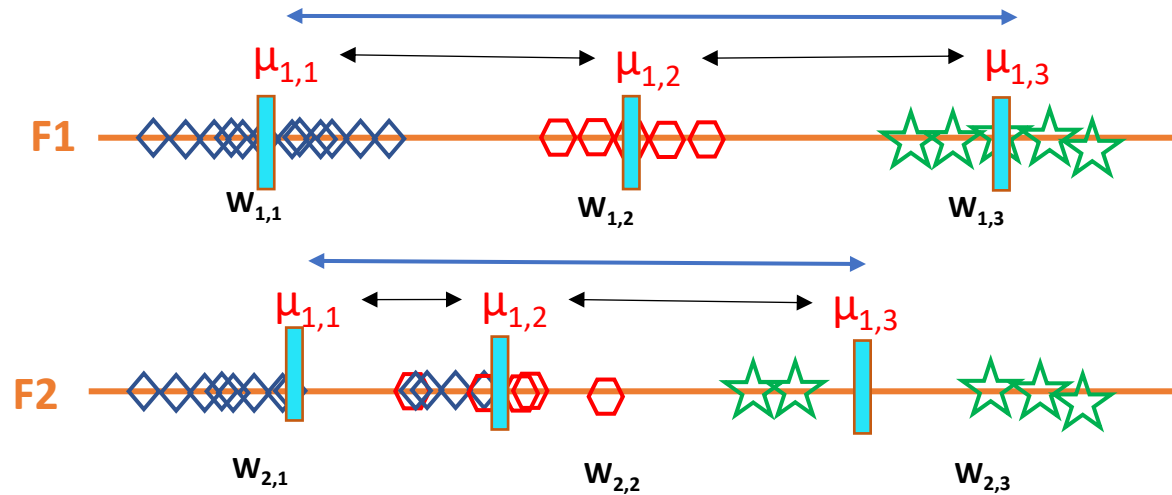
Step-2: Calculate the mean.



Proposed Method: Ir-PCA

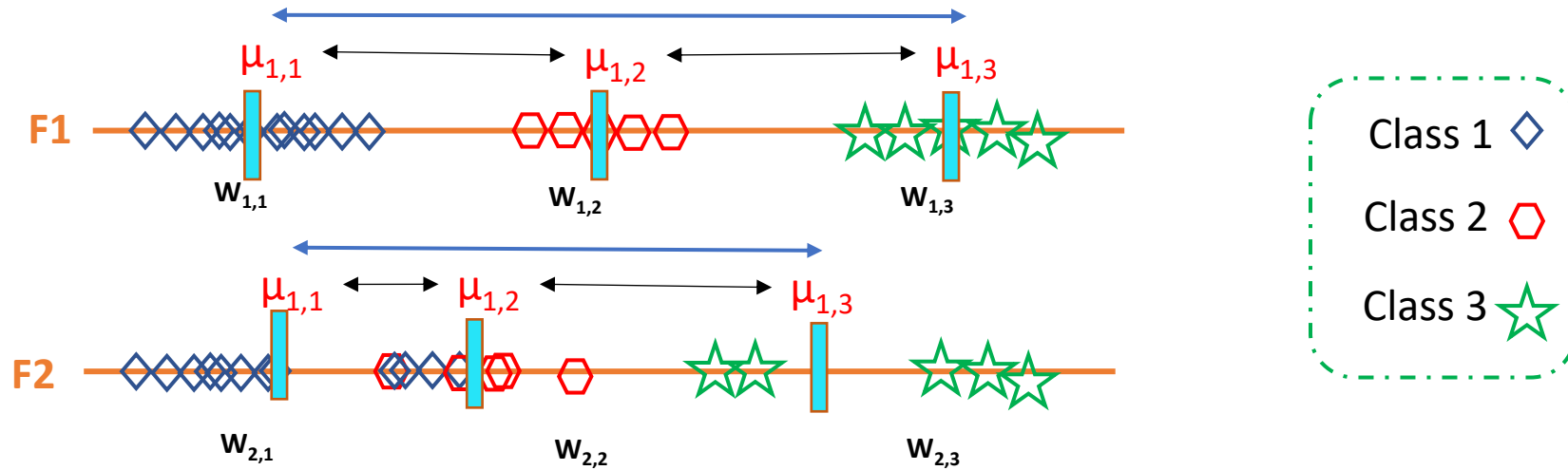
Step-3: Calculate the between classes distance.

$$T_l^d = \frac{1}{N \times (N-1)} \sum_{p,q=1}^P |\mu_{q,i} - \mu_{p,i}|, p, q = 1, 2, \dots, P, p \neq q.$$



Proposed Method: Ir-PCA

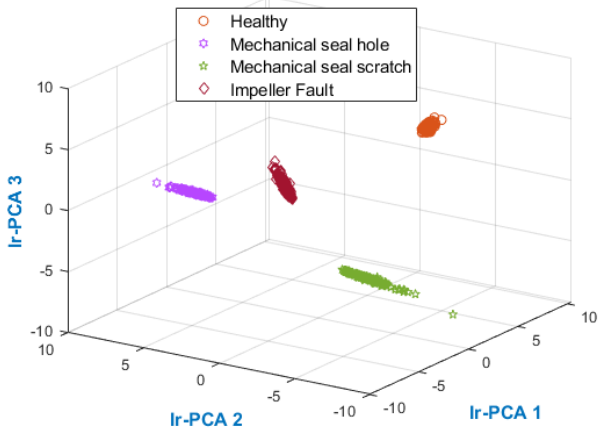
Step-3: Calculate the between classes distance.



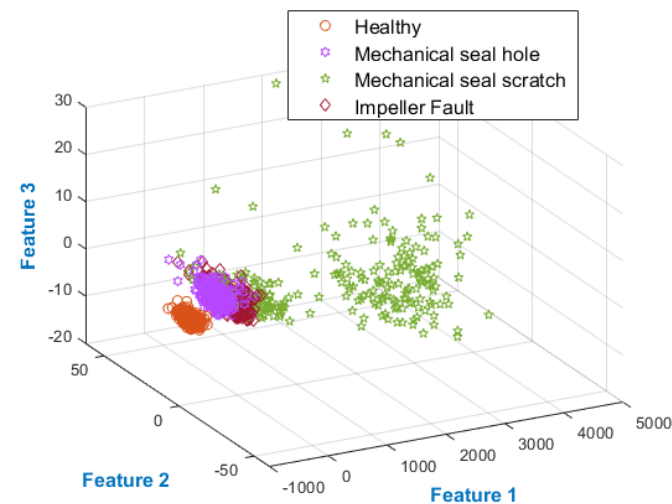
$$\text{Informative ratio } I_r = \frac{\text{Between classes distance}}{\text{within the class feature scatteredness}} \geq 0.5$$

$$\text{Ir-PCA} = \text{PCA}(\text{features}(I_r \geq 0.5)).$$

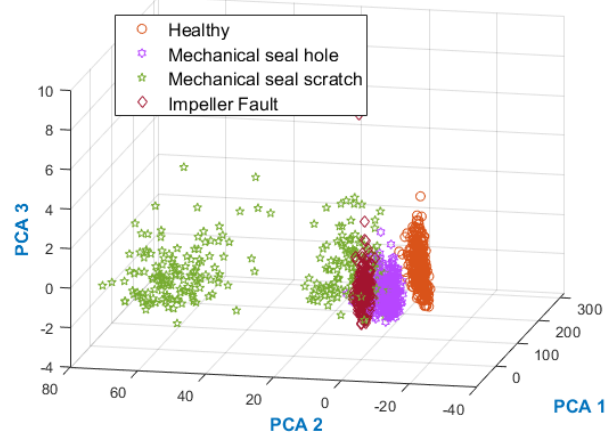
Results and Comparisons



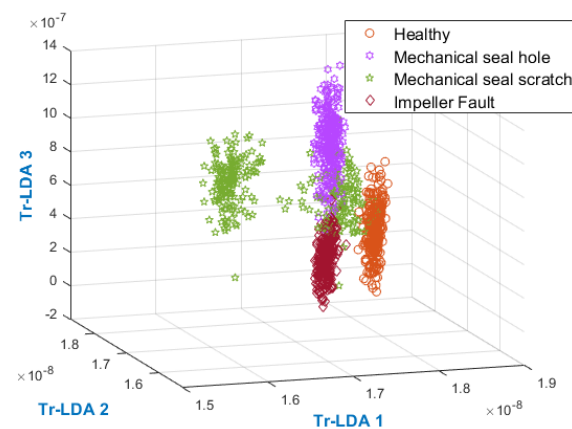
<Figure: Proposed>



<Figure: WPT-PCA-MSVM>



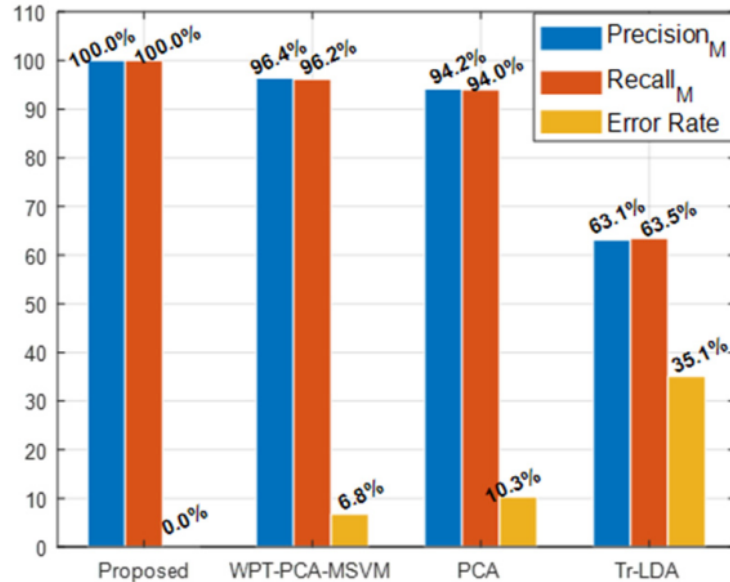
<Figure: PCA- KNN>



<Figure: Tr- LDA>

Results and Comparisons

The experiment has been performed 20 times. In each experimental trial, the training and testing sets were randomly sampled at a ratio of 8:2.



Methods	T-PR (%)				AAC (%)
	NC	MSH	MSSH	ID	
Proposed	100	100	100	100	100
WPT-PCA-MSVM	100	94.11	96.55	94.91	96.39
PCA-KNN	100	92.45	86.45	96.96	94
Tr-LDA	65.42	68.53	52.25	67.74	63.48

Average classification accuracy = 100%

$$T-PR = \frac{1}{k} \sum_{j=1}^k \left(\frac{(N_{TP}^{j,m})}{(N_{TP}^{j,m} + N_{FN}^{j,m})} \right) \times 100(\%)$$

$$AAC = \frac{1}{k} \sum_{j=1}^k \left(\sum_{m=1}^L \frac{N_{TP}^{j,m}}{N_{samples}} \right) \times 100(\%)$$

$$E_r = \frac{1}{k} \sum_{j=1}^k \left(\frac{(N_{TP}^{j,m}) + (N_{FN}^{j,m})}{(N_{TP}^{j,m}) + (N_{FN}^{j,m}) + N_{TN}^{j,m} + N_{FP}^{j,m}} \right) \times 100(\%)$$

True positive
False negatives
True Negatives
False positive

Thank You